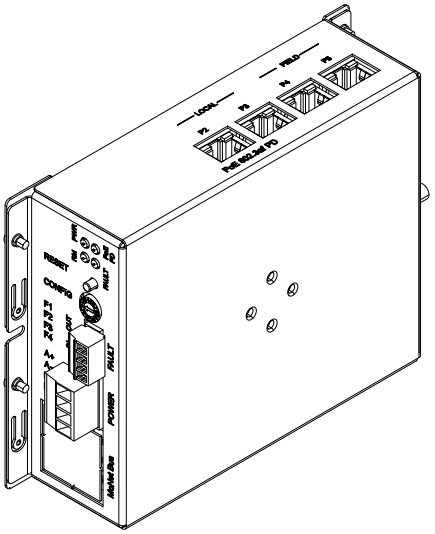
SIEMENS



FN2012-A1

Ethernet switch (modular)

Installation

Mounting

A6V10407862_f_en_--
2023-05-11

Smart Infrastructure

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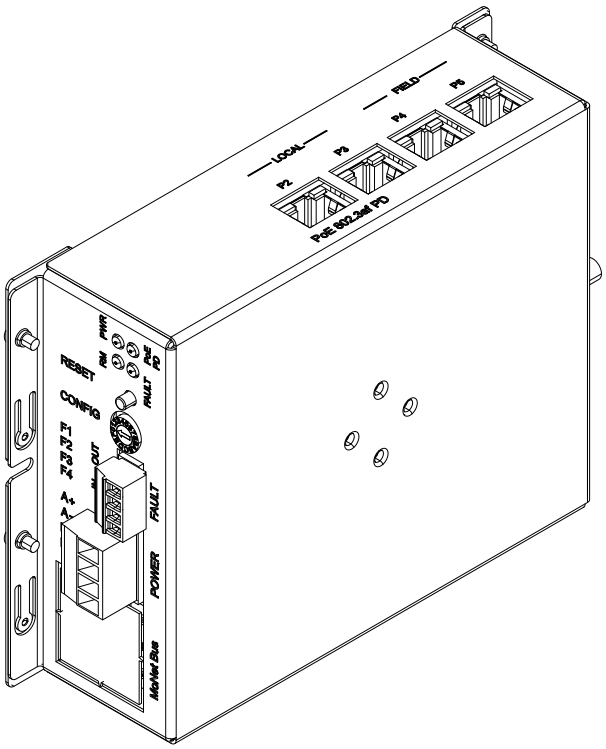
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1 Ethernet switch (modular) FN2012-A1



1.1 Description

The Ethernet switch (modular) FN2012 is intended exclusively for the operation of an Ethernet network for the physical connection to voice and VoIP components for use in fire detection.

The Ethernet switch supports system-wide signal transmission via Ethernet and an extension to the ground fault supervision for the electric Ethernet. It also supports the physical connection to voice and VoIP components.

The Ethernet switch has two slots for optional Ethernet modules. Depending on the application, the Ethernet switch is supplied via a 24 V input, via PoE using the Ethernet module (electric), or via the connection module (MoNet) FCA2031.

Intended use

- Integrated switch for Ethernet backbone in FS20/FS920 and XLS fire detection systems
- VoIP switch for Integrated Voice
- Integrated voice switch
- Operation as an Ethernet repeater
- Remote switch for external products, for example if a management station is connected

Properties

- Can be installed in approved FS20/FS920 housing
- Can be installed in approved XLS housing
- 2x Ethernet ports for local Ethernet (panel-internal)

- 2x Ethernet ports for external Ethernet (increased EMC safety and ground fault supervision)
- MoNet bus connection for transmitting the peripheral data bus signal via connection module (MoNet)
- Configurable with 16-pin rotary switch
- 1x dry contact relay output for fault signals
- Several supply options
  - Via the supply connection (redundant)
  - Via PoE with optional Ethernet module (electric)
  - Via MoNet bus
- 2x slots for optional Ethernet modules (ring ports) – not included in scope of supply
  - VN2001 Ethernet module (electric) for distances up to 330 ft/100 m
  - VN2002 Ethernet module (MM), optical Ethernet module for distances up to 13123 ft/4 km with an 11 dBm loss [1]
  - VN2003 Ethernet module (SM), optical Ethernet module for distances up to 131233 ft/40 km with a 29 dBm loss [1]

[1] The length is a guide value dependent upon the loss of the optical cable

2 Installation

2.1 Installation in FS20/FS920 housing

The Ethernet switch (modular) FN2012 can be installed in all FS20/FS920 housings (1HU and 2HU). The example below illustrates installation in a fire control panel in the housing (2HU).

If the Ethernet switch (modular) is used as a repeater, it is installed as a separate component in an external housing (1HU).

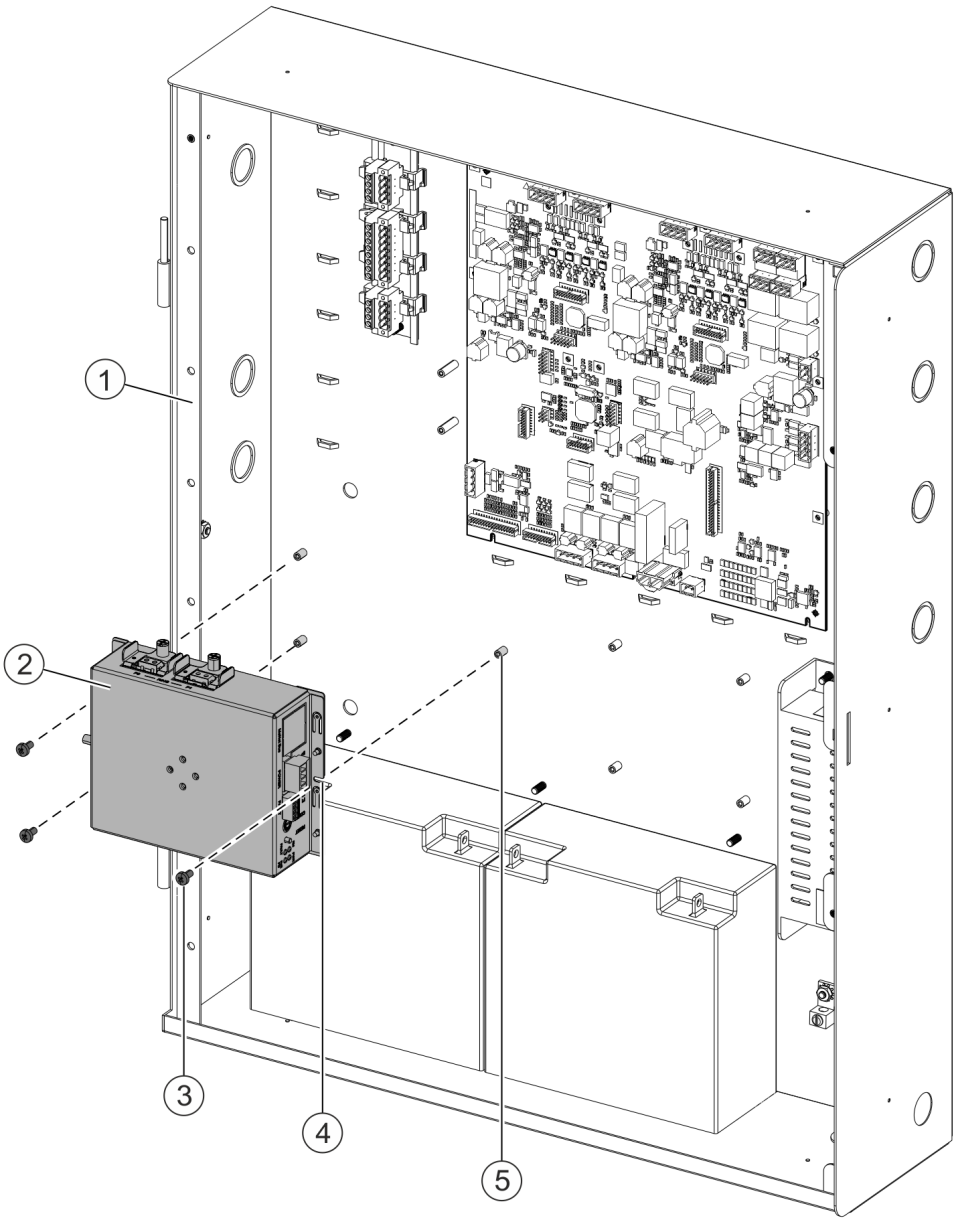


Fig. 1: Installing the Ethernet switch (modular) in housings (2HU)

1 Housing back box

2 Ethernet switch (modular) with Ethernet modules installed

3 3x fixing screw
4 Fixing slot in side tab
5 Threaded standoffs on back box

1. Position the Ethernet switch (2) as shown.
2. Inserting the 3 fastening screws (3) into the side tabs (4) and screw the Ethernet switch (2) to the threaded sleeves (5) in the back box (1).
3. Wire up the Ethernet switch according to the wiring information provided below.

2.2 Installation in XLS housings

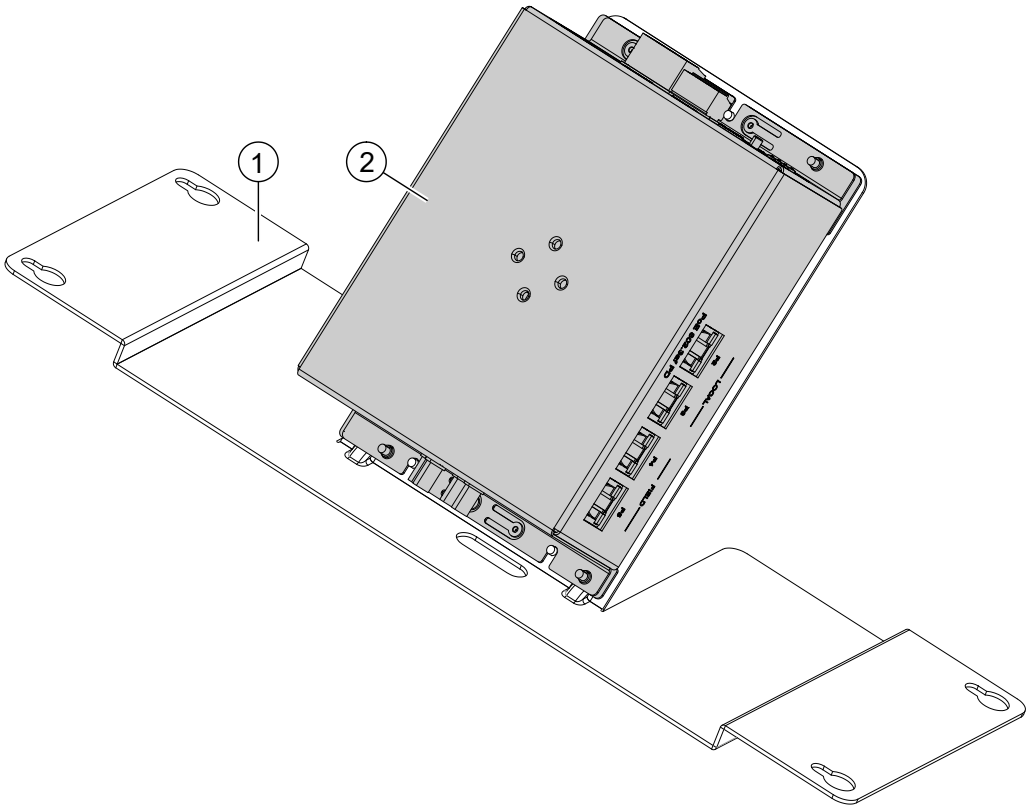


Fig. 2: Installing the Ethernet switch (modular) in XLS housings

1 Housing back box
2 Ethernet switch (modular) with Ethernet modules installed
3 3x fixing screw
4 Fixing slot in side tab
5 Threaded standoffs on back box

1. Position the Ethernet switch (2) as shown.
2. Inserting the 3 fastening screws (3) into the side tabs (4) and screw the Ethernet switch (2) to the threaded sleeves (5) in the back box (1).
3. Wire up the Ethernet switch according to the wiring information provided below.

2.3 Installation of the Ethernet modules

The FN2012 can be equipped with Ethernet modules, depending on the application. All Ethernet modules are installed in the same way.

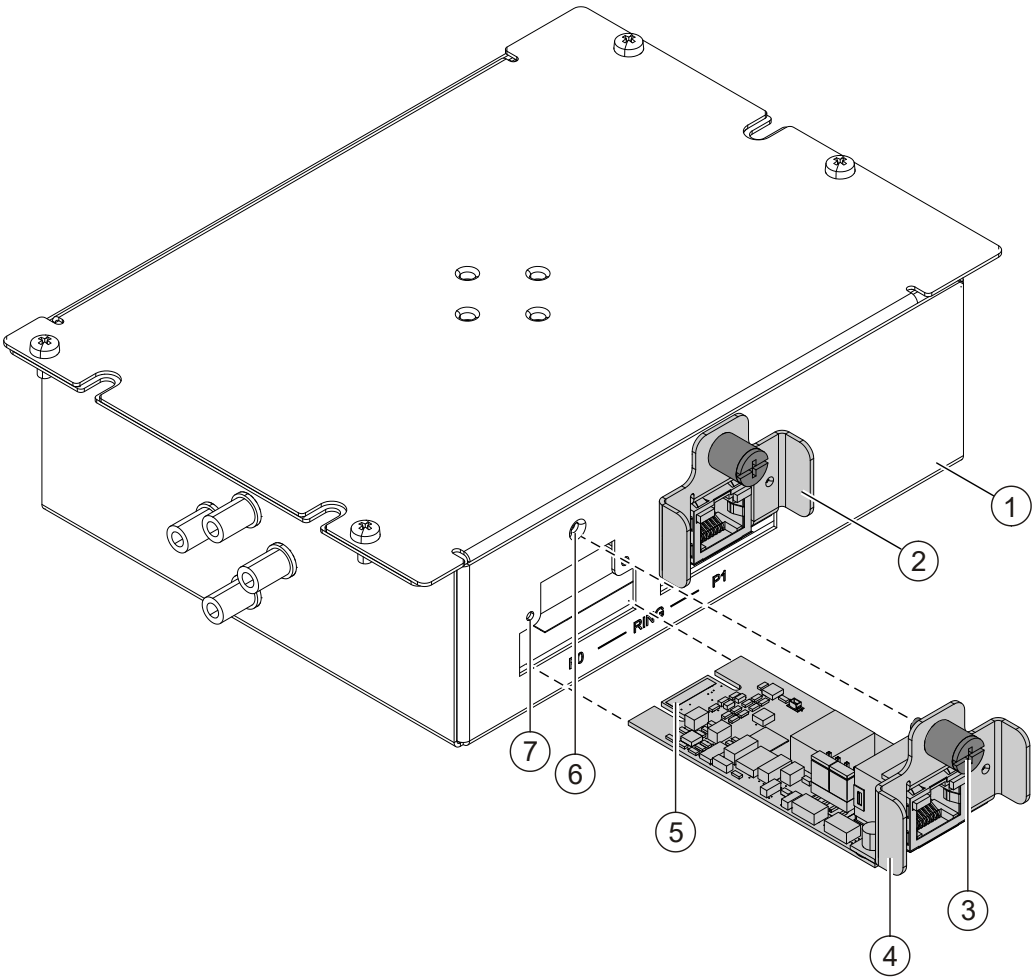


Fig. 3: Installing the Ethernet module in the FN2012

1	Ethernet switch (modular) FN2012
2	Ethernet module installed
3	Fastening screw with retention spring
4	Ethernet module
5	Plug contacts
6	Threaded hole for fastening
7	Positioning aid

Before the Ethernet module (electric) VN2001 is installed, the jumper settings must be plugged in as appropriate for how the module will be used.

1. Position the Ethernet module (4) as shown.
2. Slide the Ethernet module (4) into the opening of the option slot so that the plug contacts (5) slot into the Ethernet switch plug connector.

3. Make sure that the Ethernet module (4) slots into the positioning aids (7).
4. Use the fastening screw (3) to fix the Ethernet module in the threaded hole (6).
5. Wire up the Ethernet module according to the wiring information provided below.

See also

Jumper settings on the VN2001 [→ 20]

3 Views

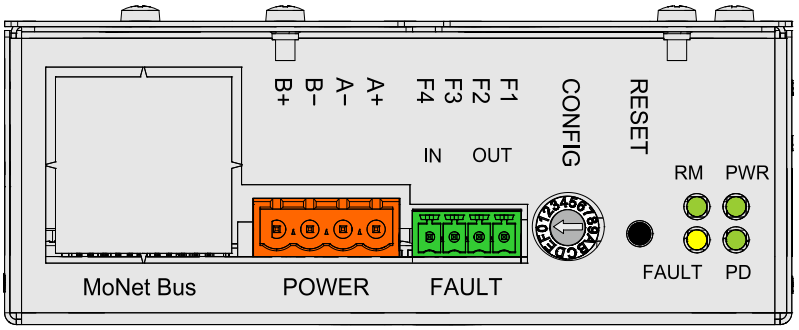


Fig. 4: View of controls and indicators FN2012

Key for the operation and indication elements

Element	Des.	Function
Connector	POWER	X301 plug connection for external supply in the same housing or room
	FAULT	X302 connector strip for fault signals: ● F1/F2, dry contact relay for fault output ● F3/F4, not connected
	MoNetBus	Knock-out sheet cover on the X201 connector strip for the MoNet bus flat cable
LEDs	RM	Redundancy manager indicator (redundancy master)
	FAULT	General fault
	PWR	Normal operation, hardware ready for operation
	PD	Not connected
Switches and keys	CONFIG	S302 rotary switch with 16 positions for pre-configured operating modes
	RESET	S301 RESET key with three functions

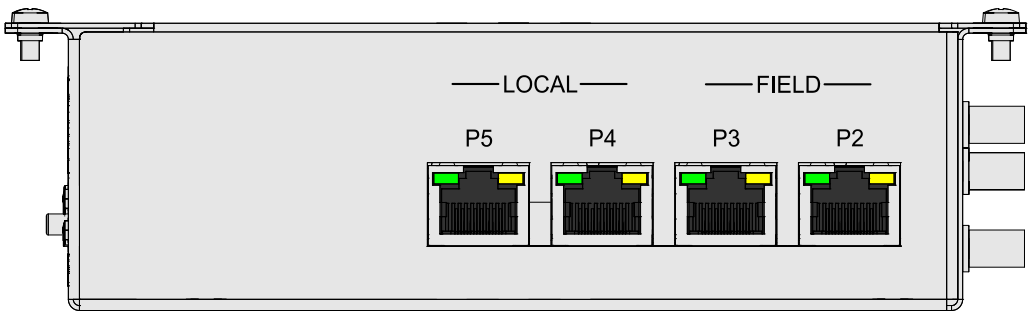


Fig. 5: View of Ethernet connections FN2012

Key for the Ethernet connections

Element	Des.	Function
Connector	P5	Internal Ethernet connection 1 (LOCAL), limited to the same housing
	P4	Internal Ethernet connection 2 (LOCAL), limited to the same housing
	P3	External Ethernet connection 1 (FIELD), limited to the same housing or the same room up to a maximum length of 20 ft/6 m
	P2	External Ethernet connection 2 (FIELD), limited to the same housing or the same room up to a maximum length of 20 ft/6 m

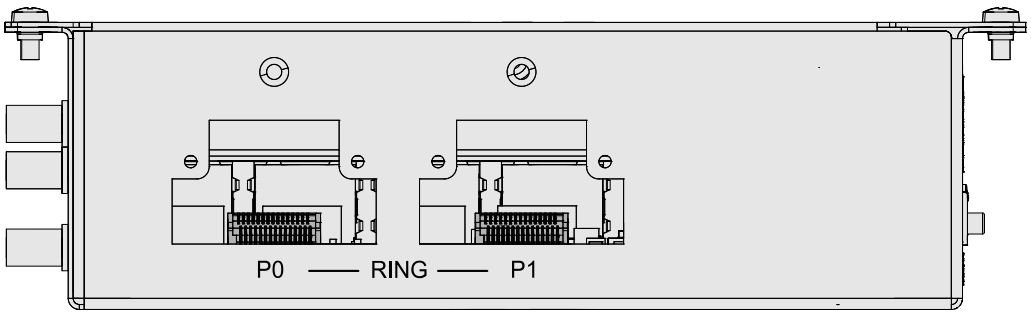


Fig. 6: View of option slots (not in use) FN2012

Key for the option slots

Element	Des.	Function
Connector	P0	RING port 0, socket X1 for the optional Ethernet module (optical or electric)
	P1	RING port 1, socket X2 for the optional Ethernet module (optical or electric)

4 Wiring

4.1 Wiring overview

The figure shows a typical application of the Ethernet switch built into an FS20 panel as well as all connections. The example shows the wiring of a panel in an optical Ethernet ring.

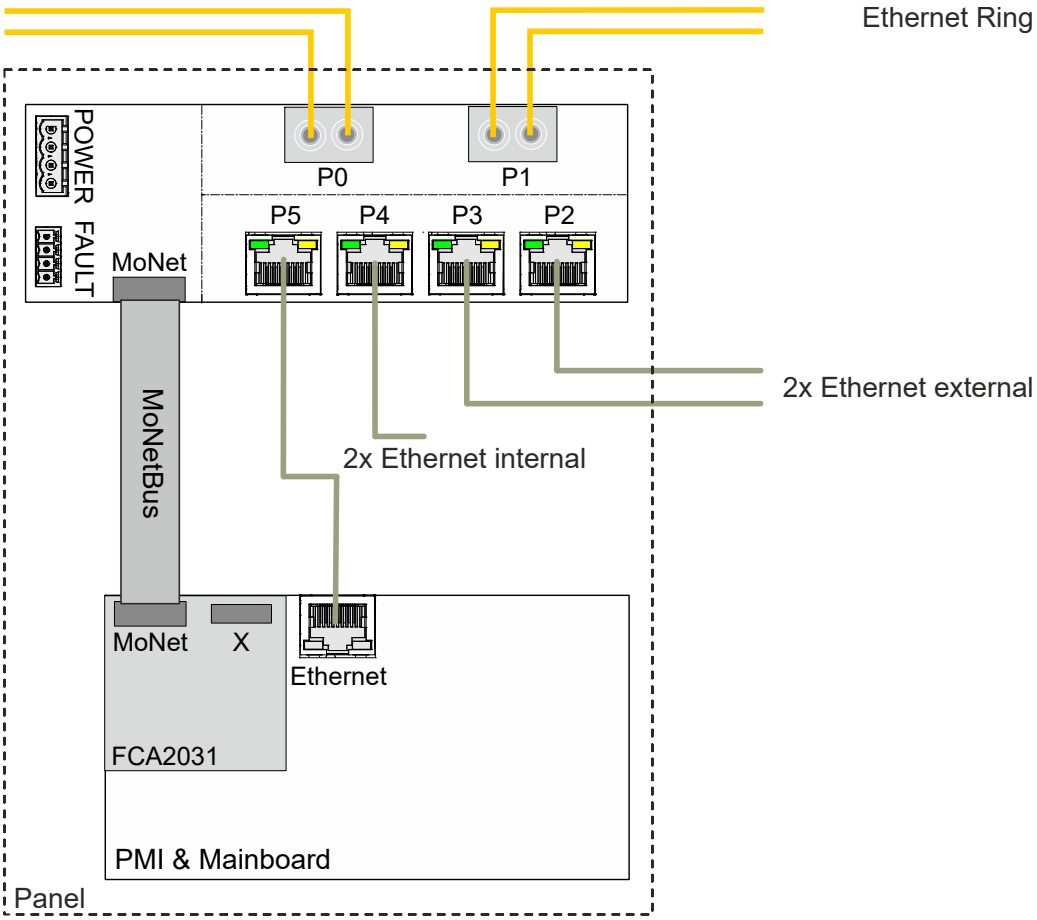


Fig. 7: Wiring of the Ethernet switch (modular) FN2012

All of the connections shown are current-limited and supervised.

Connector/Designation	Description
P0, P1	For FS20/FS920 applications: Optional modules for Ethernet ring connections for electrical or fiber-optic wiring (TP, SM, MM) For XLS applications: Optional modules for Ethernet ring connections for optical wiring only (SM/MM)
P4, P5	Panel-internal Ethernet connections
P2, P3	External Ethernet connections, e.g., for Desigo Fire Safety Works, MMS, Desigo Fire Safety View
MoNet	Connection from the PMI & mainboard to the Ethernet switch (modular) for the supply and transmission of the degraded mode and system signals (MoNetBus)

Connector/Designation	Description
FCA2031	Connection module (MoNet) in the slot on the SAFEDLINK main module. Makes the MoNet bus available.
X	Not used
Panel	Approved Siemens panel in which the Ethernet switch (modular) is being installed

The electrical Ethernet cable for connections P2...P5, as well as the connections via the Ethernet module (electric) VN2001 to connections P0 and P1 must be at least 6 ft/ 2 m long.

External Ethernet connections P2 and P3 are limited to the same room and have a maximum length of 20 ft/6 m.

The internal Ethernet connections P4 and P5 are limited to the same housing.

4.2 General wiring FS20/FS920

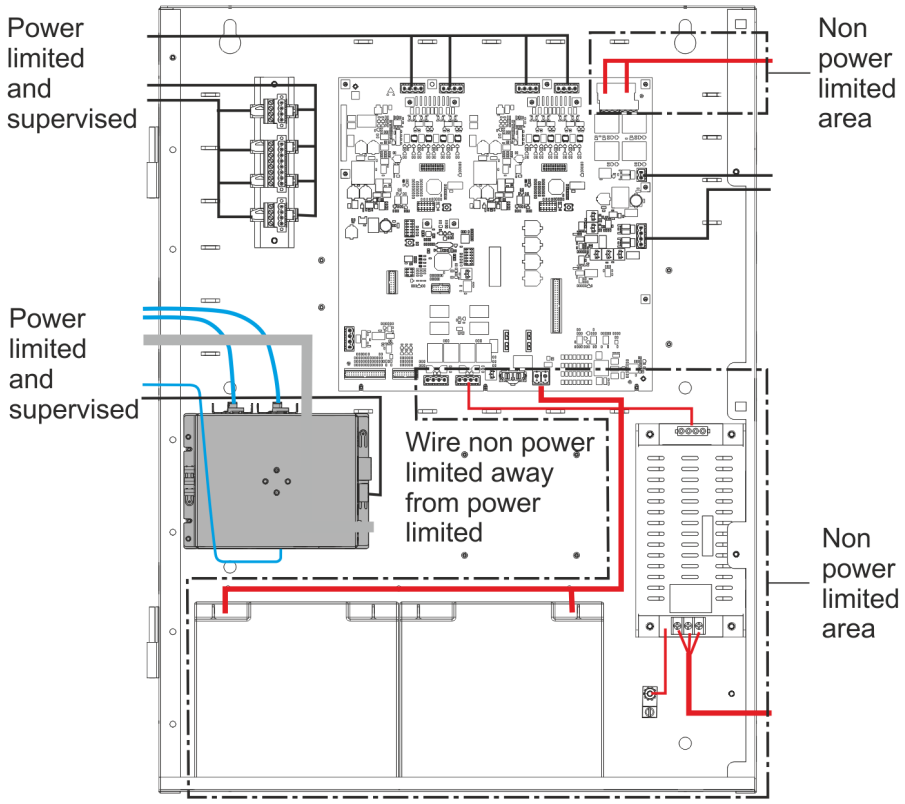


Fig. 8: View of cable routing

Supply and battery connection

The wiring must be routed as far away as possible from the installed components and the non-current-limited cables and fixed in place appropriately.

Cable guides in the housing

Current-limited and non-current-limited cables must not be routed together through the same cable entry.

4.3 General wiring XLS

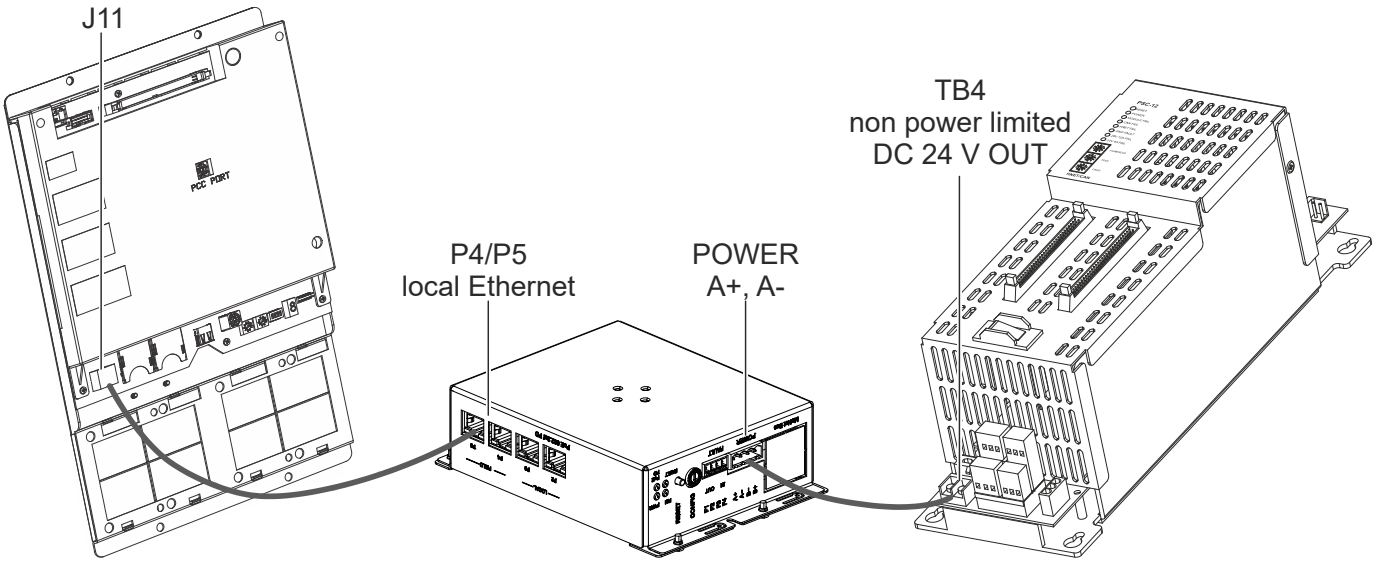


Fig. 9: Wiring overview of XLS components

- The Ethernet switch (modular) is supplied via the PSC-12 power supply, connection TB4.
- The internal Ethernet connection is established between port 4 or port 5 on the Ethernet switch (modular) and the PMI-2, connection J11.

4.4 MoNet bus connection

Preparing the Ethernet switch (modular) for MoNet bus connection

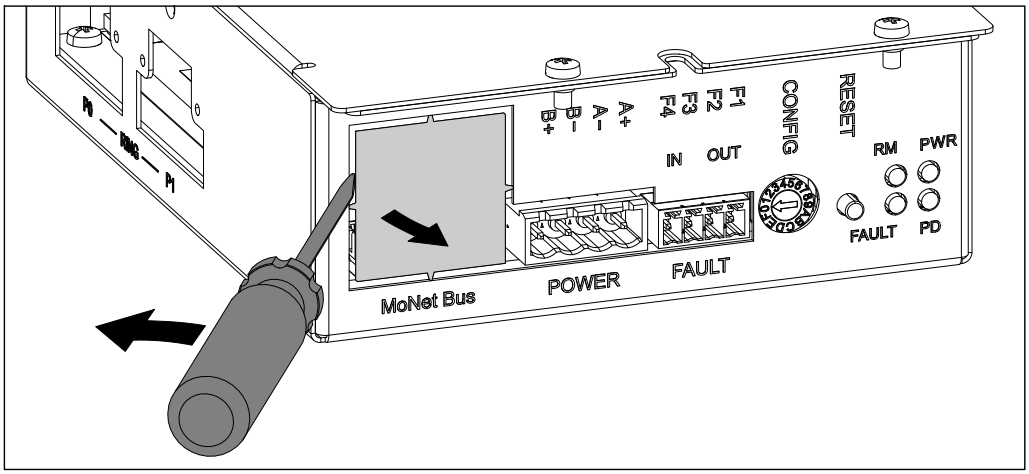


Fig. 10: Breaking out the MoNet bus cover plate

◈ Use a screwdriver to carefully remove the cover plate for the MoNet connector as shown in the illustration.

⇨ Levering the screwdriver in the direction shown by the arrow causes the cover plate to come loose at the four predetermined breaking points.

Ribbon cable connection for MoNet bus

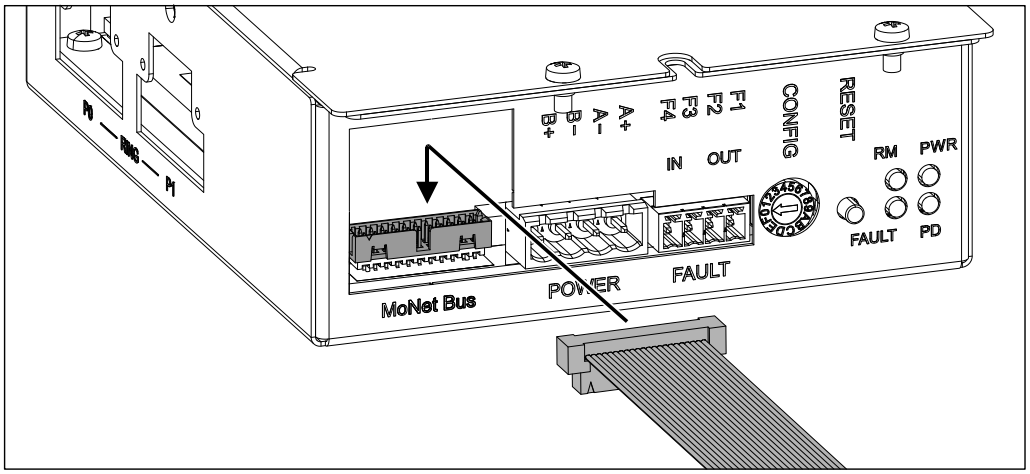


Fig. 11: Connecting the MoNet bus ribbon cable

- Connect the 1.35 m/4.43 ft long ribbon cable for the MoNet bus as shown on the Ethernet switch (modular).

4.5 Ethernet switch in the Ethernet network

The basic circuit diagram below shows the possible applications of the Ethernet switch in a redundant Ethernet ring network.

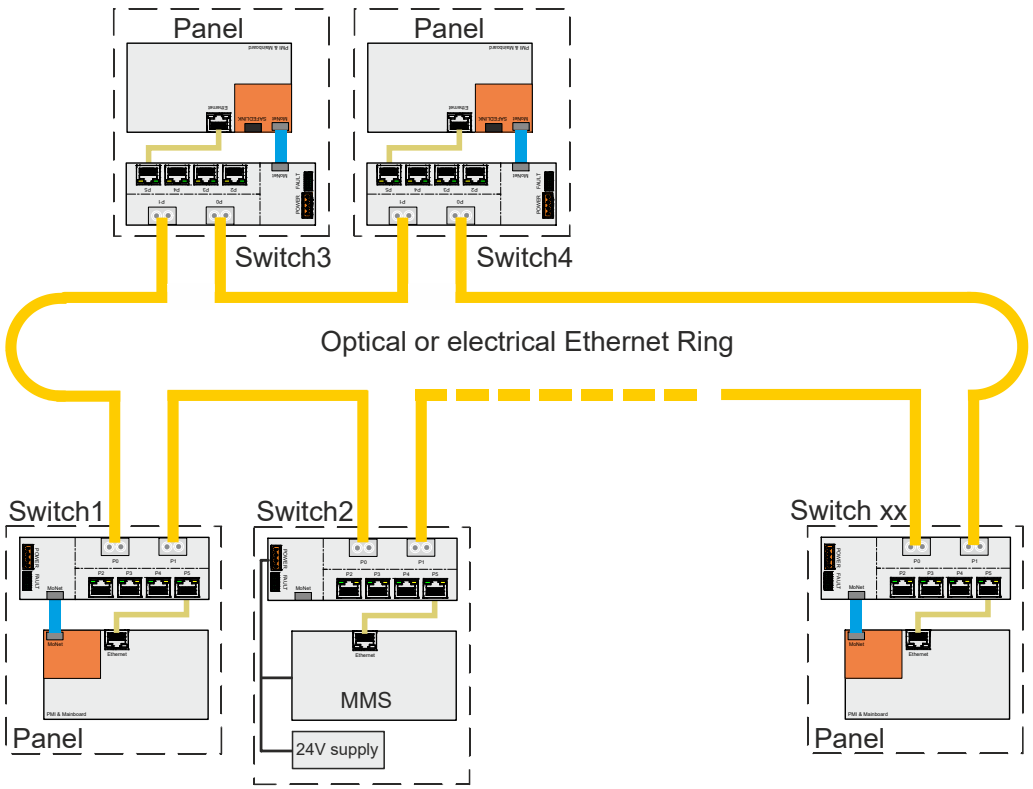


Fig. 12: Ethernet switch in a redundant Ethernet ring network

There are 3 types of Ethernet ring wiring, depending on the Ethernet module in use:

- Electric Ethernet with copper wiring via 10/100BaseT connections (RJ-45 connector) with a VN2001 module – max. 330 ft/100 m long
- Optical Ethernet with a fiber-optic 100BaseFX connection in multi-mode version with a VN2002 module – max. 13100 ft/4 km long
- Optical Ethernet with a fiber-optic 100BaseFX connection in single-mode version with a VN2003 module – max. 131000 ft/40 km long

The type of ring wiring between each switch is independent and can be selected at will. A maximum of 32 switches are allowed for each Ethernet ring.

The switch built into an FS20 panel is supplied via the MoNetBus from the PMI & mainboard. If the switch is built into another housing or device, it is supplied with power via a 24 V current-limited and supervised power supply that has been approved for use in fire detection.

Ring wiring with Ethernet module (electric) VN2001

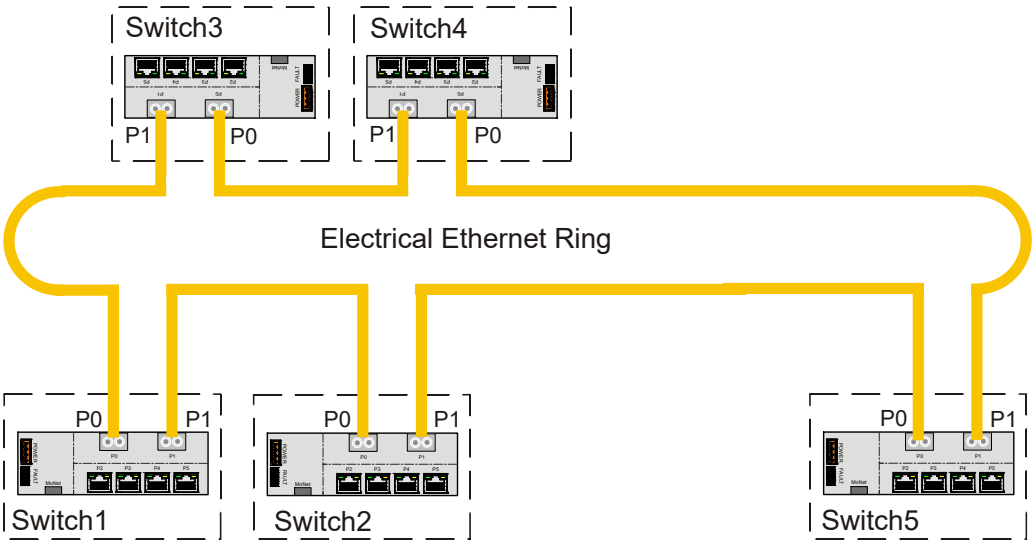


Fig. 13: Example of ring wiring with electrical Ethernet (simplified representation)

Jumper settings on the VN2001 for electrical Ethernet:

Switch	VN2001 in port P0	VN2001 in port P1
1	open	Ground fault supervision
2	open	Ground fault supervision
3	open	Ground fault supervision
4	open	Ground fault supervision
5	open	Ground fault supervision

The minimum cable length for an electrical 10/100BaseT cable on the VN2001 is 10 ft/3 m. Shorter lengths are only permitted within the same housing, or if the connection does not need to be supervised in accordance with the UL864 standard.

4.6 Ethernet switch as a repeater

The Ethernet switch (modular) can also be used as a repeater in the electric Ethernet. For each repeater, the distance of the electric Ethernet connection increases by 330 ft/ 100 m. The repeater is supplied with power via PoE from the installed Ethernet module (electric) VN2001.

The arrows in the figures symbolize the function of the ground fault supervision and the supply via PoE from the Ethernet module (electric).

The settings for PoE and ground fault supervision are made by plugging jumpers into slots P0 and P1 on the corresponding Ethernet module (electric).

One repeater

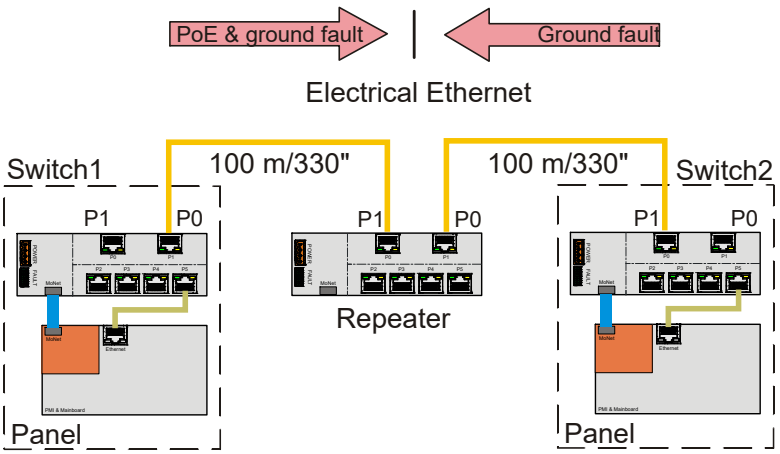


Fig. 14: 1x Ethernet switch (modular) as a repeater in the electric Ethernet

Jumper settings on the VN2001 with one repeater:

Switch/repeater	VN2001 in port P1	VN2001 in port P0
Switch 1	Not connected	PoE
Repeater	PoE	Neutral position
Switch 2	Ground fault supervision	Not connected

Two repeaters

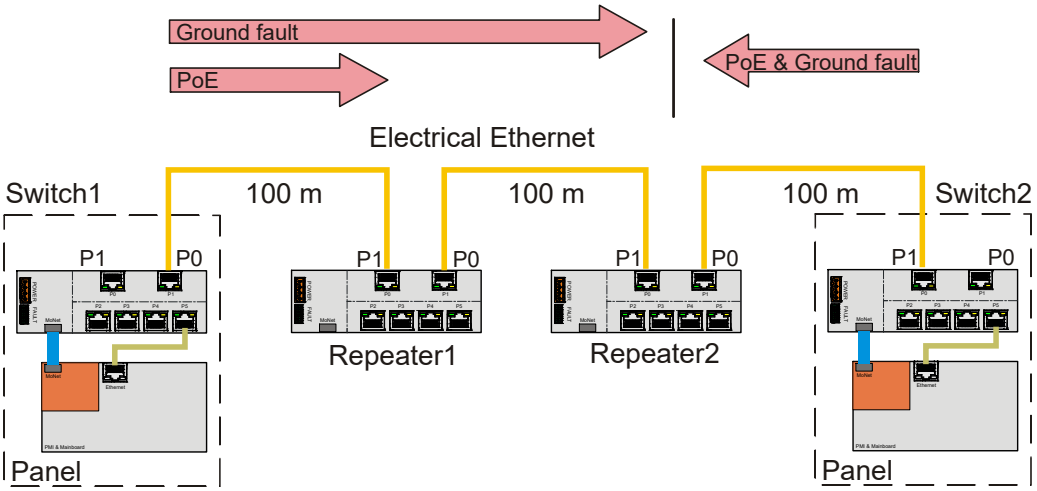


Fig. 15: 2x Ethernet switch (modular) as a repeater in the electric Ethernet

Jumper settings on the VN2001 with two repeaters:

Switch/repeater	VN2001 in port P1	VN2001 in port P0
Switch 1	Not connected	PoE
Repeater 1	PoE	Ground fault supervision
Repeater 2	Neutral position	PoE
Switch 2	PoE	Not connected

Guidelines

- Max. two repeaters between two nodes.
- Electric Ethernet distance between two nodes maximum 300 ft/100 m.
- Repeater power supply via 'Power over Ethernet' (PoE) from the VN2001.
- For each switch (node), 1 repeater can be supplied with power.
- The switch carries out ground fault supervision as far as the repeater.
- If there are two repeaters, the first one transfers ground fault supervision to the second repeater.

5 Jumper settings on the VN2001

The slots the jumpers are plugged into on the Ethernet module (electric) differ depending on how the module will be used. The table below shows the different settings for the corresponding function.

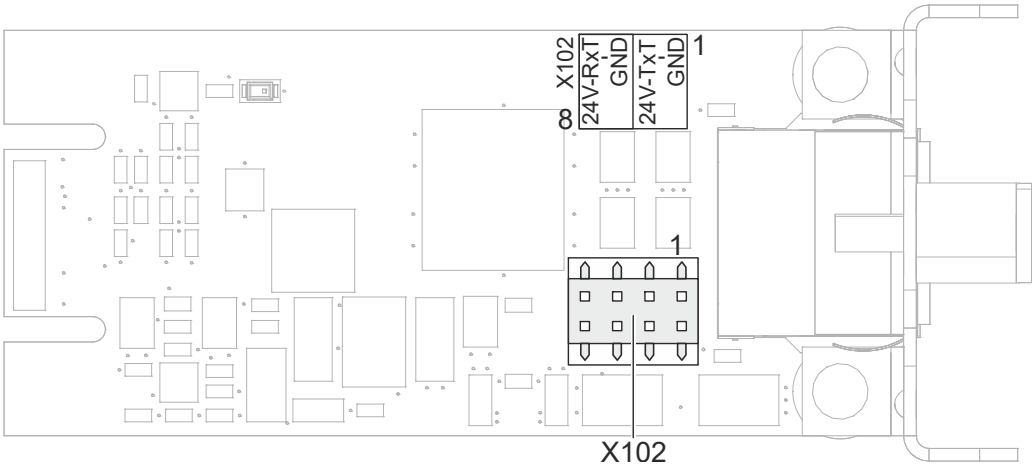
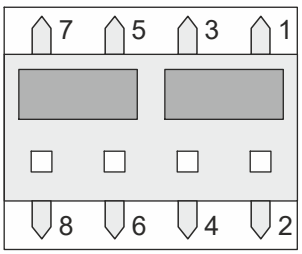
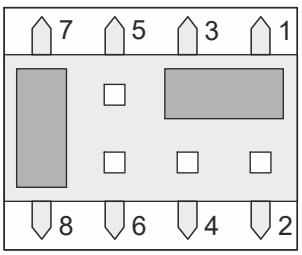
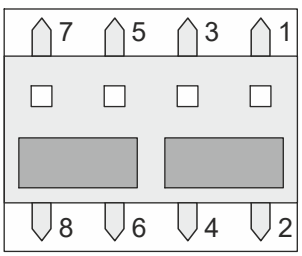
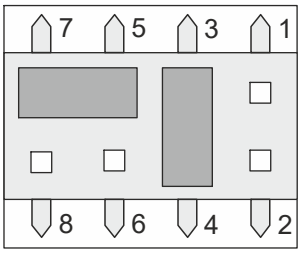


Fig. 16: View of VN2001 with jumper strip

X102 jumper strip

Jumper setting on X102	Function
	Ground fault supervision activated
	PoE activated
	Neutral position, function deactivated

Jumper setting on X102	Function
	PoE for cables crossed at one end

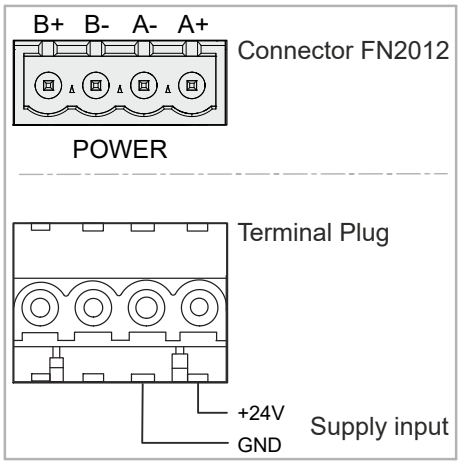
6 Pin assignments

6.1 'POWER' connector strip

The 'POWER' connector is only needed if the supply is not possible via the MoNetBus or PoE. This is generally only the case if the Ethernet switch (modular) is installed remotely in another device. The power supply must be a DC 24 V power supply featuring regulated and power-limited output that has been approved for fire alarm signal applications.

Designation	Description
A+	Supply input 1 (+DC 24 V)
A-	Common supply input 1 (DC 0 V)
B-	Not connected
B+	Not connected

Admissible cable cross-section: AWG 24...AWG 12 / 0.2...2.5 mm²



6.2 'FAULT' connector strip

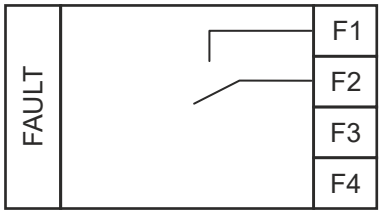
The fault output is a dry contact relay and, depending on the configuration, it can be used to report fault states by means of contact interruption. When the FN2012 is switched off, the relay contact is deactivated (opened).

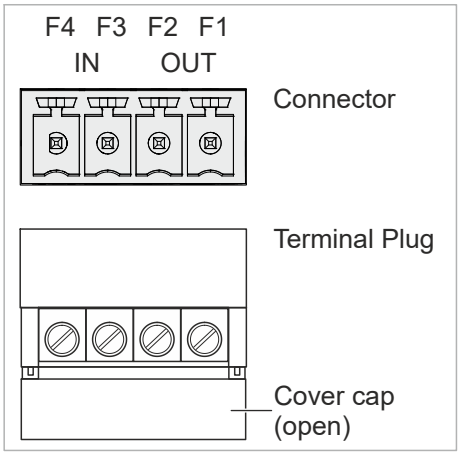
The following fault messages can be configured via the web interface:

- 'Fatal fault': Ethernet switch (modular) malfunction
- 'Ring fault': Ring faults are only displayed if the Ethernet switch (modular) has the function of the 'redundancy manager'
- 'Ring port fault': 'Link down' state of a ring port

Pin	Designation	Description
F1	OUT	Dry contact relay 1 (common, normally open)
F2	OUT	Dry contact relay 2 (common, normally open)
F3	IN	Not connected
F4	IN	Not connected

Admissible cable cross-section: AWG 24...AWG 12 / 0.2...2.5 mm²





7 Indicators

LED	Color	Function	Condition	Meaning
PWR	Green	Power LED	Lit up	• Normal operation
			Not lit up	• No power supply
FAULT	Yellow	General fault	See separate FAULT LED table	
RM	Green	Redundancy manager	See separate RM LED table	
PD	Green	Not connected	--	No function
			--	No function
P2	Green/yellow	Ethernet status LEDs	Green ON	• Link up
P3			Green OFF	• Link down
P4			Yellow flashing	• Data communication active
P5			Yellow does not light up	• No data communication

Table 1: Status LED

Function indicators for the FAULT LED

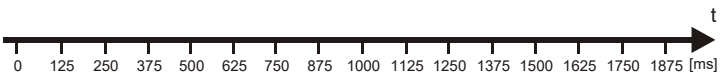
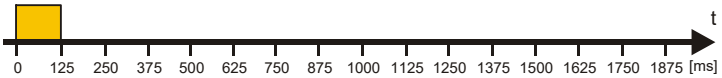
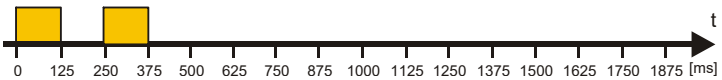
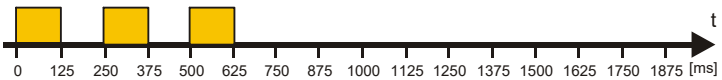
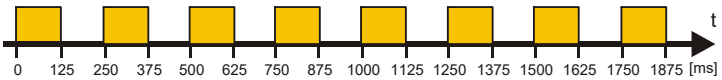
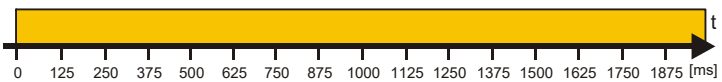
Function indicator	Meaning	FAULT LED flashing pattern (every 2 s)
Normal Operation	Normal operation	
FW Update Mode	Ready for firmware update	
FW Update Running	Ethernet switch is loading firmware	
FW Update Failed	FW update could not be completed, network connection may have been interrupted	
Checksum Failed	Incorrect firmware checksum	
Fatal FAULT	Ethernet switch is generating some other fault	

Table 2: FAULT LED flashing pattern

Function indicator	Possible troubleshooting steps
FW Update Failed	• The Ethernet switch attempts to re-establish the connection in order to load the FW. • Restart the FW update via Desigo Fire Safety Works.
Checksum Failed	• Restart the FW update via Desigo Fire Safety Works.

Function indicator	Possible troubleshooting steps
Fatal FAULT	• Perform a reset. If that does not work, there is a hardware fault. • Replace the Ethernet switch. Check whether any other fault messages are being displayed in the FS20 system.

Table 3: Troubleshooting

Function indicators for the RM LED (redundancy manager)

Within each MRP [1] ring, there has to be a main Ethernet switch. All the other Ethernet switches must perform the role of clients.

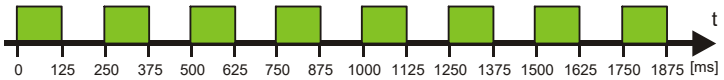
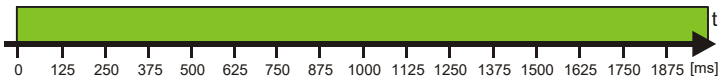
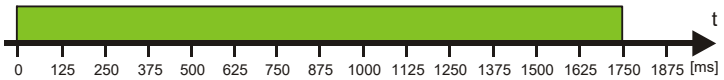
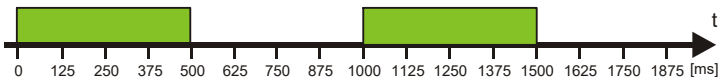
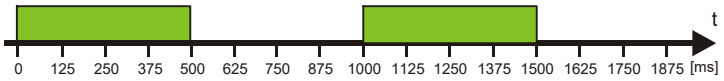
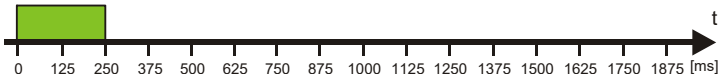
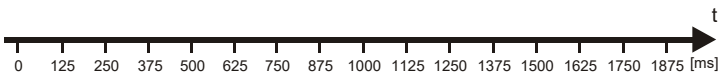
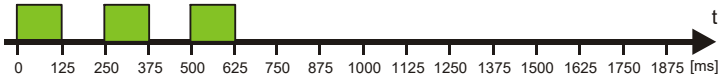
Function indicator	Meaning	RM LED flashing pattern (every 2 s)
Search Manager	Ethernet switch is being initialized System is looking for MRP manager Topology has changed because Ethernet ring participants have been added or removed	
Manager	Ethernet switch has been configured as the manager and is performing the ring manager role	
Claim Manager	Ethernet switch is acting as the manager but has not been configured to perform this role	
Manager Ring open	There is a break in the Ethernet ring at one or more locations	
Claim Manager Ring open		
Double manager	Two Ethernet switches have been configured as the manager	
Client	Ethernet switch has been configured as a client	
Unidirectional Data Link Fault	In one section of the MRP ring, the connection is only running in one direction due to incorrect cabling or a hardware fault	

Table 4: RM LED flashing pattern

[1] MRP: Media redundancy protocol

Function indicator	Possible troubleshooting steps
Claim Manager	• Adapt configuration. A ring manager has to be configured within each MRP ring.

Function indicator	Possible troubleshooting steps
Ring open	● Check the MRP ring. There is a break in the ring at one or more of the panels. ● Check whether cable is defective. ● Ring port LEDs which are switched off show open ports – Defective/unplugged cable.
Double manager	● Adapt configuration. Only one Ethernet switch is allowed to be configured as the ring manager.
Unidirectional Data Link Fault	● Check the cabling of all optical fiber sections ● Replace each Ethernet switch in turn until you identify the defective device.

Table 5: Troubleshooting

8 Adjustment elements

Reset button

The reset button has a number of functions depending on how long it is pressed for.

RESET button pressed for …	FAULT LED status	Function/result
<5 s	Not lit up	**Hardware reset**: The FN2012 is restarted immediately after pressing the reset button.
5…10 s	Lit up	**Factory reset**: The FN2012 is restored to the factory settings.
>10 s	Lights up and goes out as soon as the FN2012 is in update mode	The FN2012 is in **firmware update mode**. The reset key can be released.

Rotary switch for device configuration

Configuration via the rotary switch is required for an Ethernet switch that is not integrated into a panel. As an integrated Ethernet switch, the rotary switch must be set to '0'.

By using the rotary switch for device configuration, the Ethernet switch can be set to predefined network addresses or operation modes.

To accept a new configuration, the switch must be in a de-energized state when the setting is made or a hardware reset must be performed afterward.

Switch position	IP address/operating mode	Sub-net mask/operating mode
0	Configured via web interface or MoNet bus Factory setting: 0.0.0.0	Configured via web interface or MoNet bus Factory setting: 255.255.248.0
1	192.168.99.241	255.255.248.0
2	192.168.99.242	255.255.248.0
3	192.168.99.243	255.255.248.0
4	192.168.99.244	255.255.248.0
5	192.168.99.245	255.255.248.0
6	192.168.99.246	255.255.248.0
7	192.168.99.247	255.255.248.0
8	192.168.99.248	255.255.248.0
9	192.168.99.249	255.255.248.0
A	192.168.99.250	255.255.248.0
B	192.168.99.251	255.255.248.0
C	192.168.99.252	255.255.248.0
D	Reserved (0.0.0.0)	Reserved (255.255.255.255)
E	Repeater mode	Repeater mode

Switch position	IP address/operating mode	Sub-net mask/operating mode
F	192.168.200.4	255.255.255.0

9 Technical data

Supply	Operating voltage	DC 24 V
	Standby power consumption	95 mA @24 V
	Built-in power consumption, supply via MoNetBus or AUX power	Max. 380 mA @ 24 V
	Power consumption via PoE	Max. 185 mA @ 24 V
Ethernet interfaces	Sockets	4x RJ45, 10/100 Mbit/s
	Length of line	● P4, P5 'Local', limited to same housing but must be min. 2 m ● P2, P3 'Field', limited to same room but must be min. 2 m, max. 6 m
Connectors	External supply input	4-pin plug-in terminal
	Fault terminal	4-pin plug-in terminal
	● Fault output 'F1', 'F2'	Dry contact relay (common) Switching current max. 1 A, resistive Switching voltage max. DC 30 V
	MoNetBus	22-pin flat-cable connector
	External Ethernet connections 'P2', 'P3'	● Additional voltage surge protection ● Increased EMC protection ● Ground fault supervision
	Circuits	● All connections are power limited ● All connections – except local and MoNet connections – are supervised
Module connections (optional)	Ethernet module (electric) VN2001	RJ45 connection (TP), cable length max. 330 ft/100 m
	Ethernet module (MM) VN2002	LC connector, multi-mode fiber-optic ● Fiber length 62.5/125 µm: 13123 ft/ 4 km with an 11 dB loss [1] ● Fiber length 50/125 µm: 6561 ft/2 km with a 7.5 dB loss [1]
	Ethernet module (SM) VN2003	LC connector, single-mode fiber-optic: ● Fiber length 9/125 µm: 131233 ft/ 40 km with a 29 dB loss [1]
Mechanical data	Dimensions (W x H x D), with cover, without optional modules	165 x 121 x 43 mm
	Weight	560 g/1.2 lbs
Ambient conditions	For indoor use in dry rooms only	
	Operating temperature	120+/-3 °F to 32+/-3 °F
	Rel. humidity during operation	93+/-2 % at a temperature of 90+/-3 °F
	Protection category	IP30

[1] The length is a guide value dependent upon the loss of the cable used, the loss of the connector types used, and the transmission losses that are present. The corresponding values must be taken from the respective data sheets.

Sample calculation

- Optical budget for VN2002: 11 dB
- Cable loss: 2.5 dB/3280 ft / 2.5 dB/km
- Plug loss: 1.0 dB
- Transmission losses: 0.5 dB

Length = (optical budget – 2x plug loss – transmission losses) / (cable loss per km)

Length in km = (11 dB – 2 x 1 dB – 0.5 dB) / 2.5 dB per km = 3.4 km

Length in ft = (11 dB – 2 x 1 dB – 0.5 dB) / 2.5 dB/3280 ft = 9842 ft

10 FCC Statement

⚠ WARNING	
	**Installation and usage of equipment is not in accordance with instructions manual** Radiation of radio frequency energy Interference to radio communications ● Install and use equipment in accordance with instructions manual. ● Read the following information.

This equipment generates, uses, and can radiate radio frequency energy and if not installed and used in accordance with the instructions manual, may cause interference to radio communications.

It has been tested and found to comply with the limits for a Class A computing device pursuant to Part 15 of FCC Rules, which are designed to provide reasonable protection against such interference when operated in a commercial environment.

Operation of this equipment in a residential area is likely to cause interference in which case the user at his own expense will be required to take whatever measures may be required to correct the interference.

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